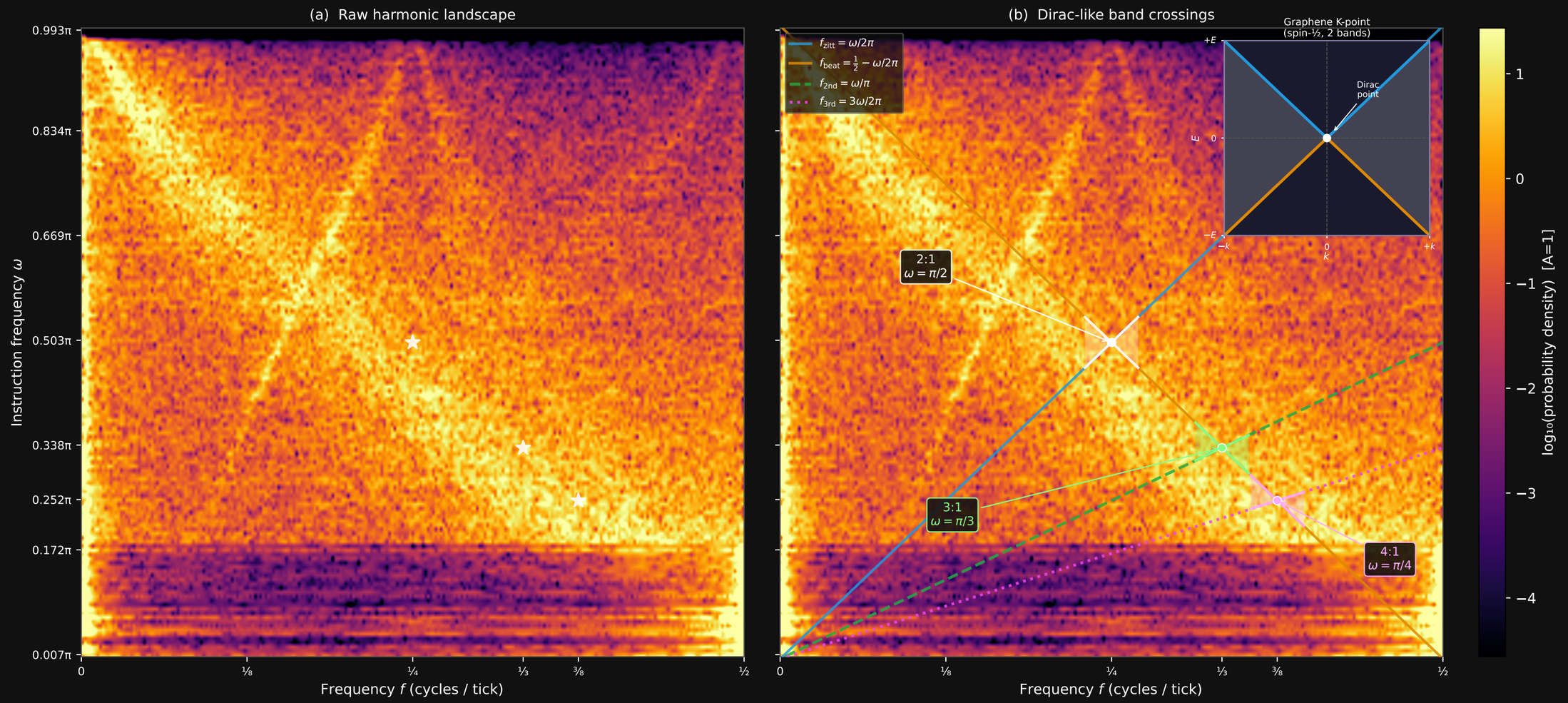
(a) Raw harmonic landscape
(b) Dirac-like band crossings
Instruction frequency ω
0.993π
0.834π
0.669π
0.503π
0.338π
0.252π
0.172π
0.007π
0
⅛
¼
⅓
⅜
½
Frequency f (cycles / tick)
$f_{\text{zitt}} = \omega/2\pi$
$f_{\text{beat}} = \frac{1}{2} - \omega/2\pi$
$f_{\text{2nd}} = \omega/\pi$
$f_{\text{3rd}} = 3\omega/2\pi$
2:1
$\omega = \pi/2$
3:1
$\omega = \pi/3$
4:1
$\omega = \pi/4$
Graphene K-point
(spin-½, 2 bands)
Dirac point
+E
−E
−k
0
k
+k
log₁₀(probability density) [A=1]
1
0
−1
−2
−3
−4